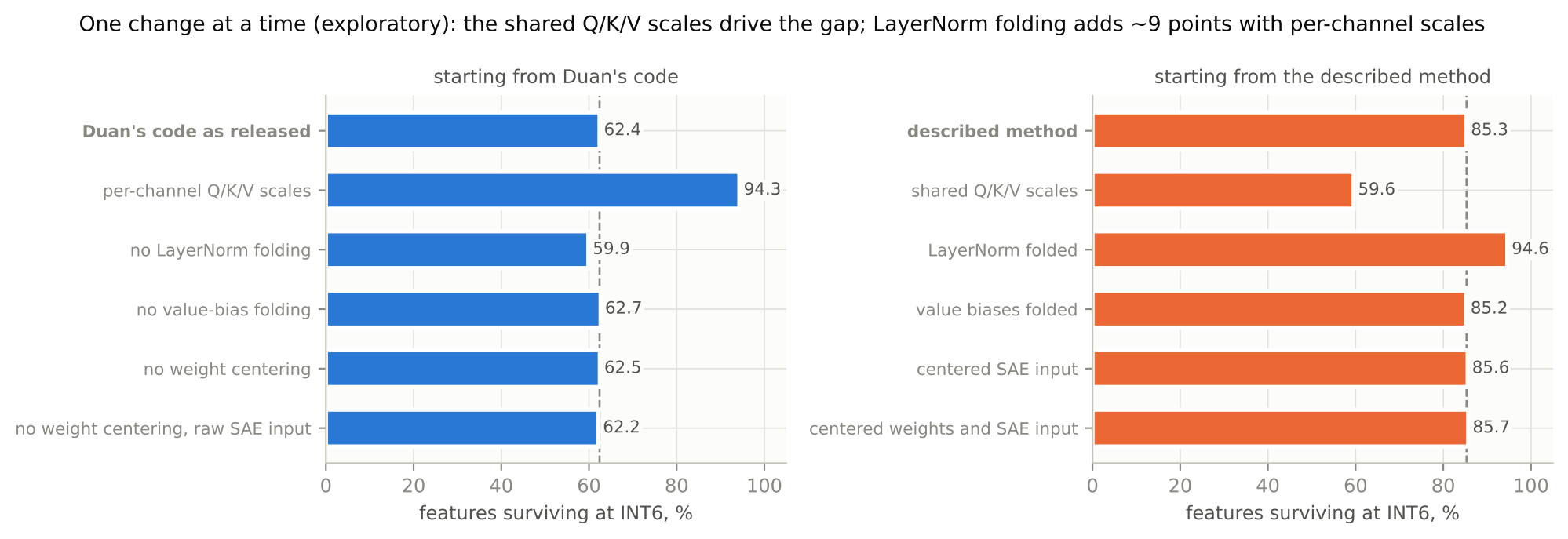
One change at a time (exploratory): the shared Q/K/V scales drive the gap; LayerNorm folding adds ~9 points with per-channel scales
starting from Duan's code
Duan's code as released
62.4
per-channel Q/K/V scales
94.3
no LayerNorm folding
59.9
no value-bias folding
62.7
no weight centering
62.5
no weight centering, raw SAE input
62.2
0
20
40
60
80
100
features surviving at INT6, %
starting from the described method
described method
85.3
shared Q/K/V scales
59.6
LayerNorm folded
94.6
value biases folded
85.2
centered SAE input
85.6
centered weights and SAE input
85.7
0
20
40
60
80
100
features surviving at INT6, %